# Customer Behavior Analysis

Alfido Tech Internship Project | Dataset-driven analysis and recommendations

## 1. Executive summary

The primary dataset contains **250,000** transaction rows and **49,661** unique customers, covering 01 Jan 2020 to 13 Sep 2023. Recorded total purchase amount is **681,346,299** (dataset currency is not specified), with an average transaction amount of **2,725.4**. The supplied customer-level churn label is **20.0%**. RFM analysis indicates 51.1% of customers had no purchase in the 180 days before the final dataset date; this is an inactivity proxy, not a confirmed churn rate.

## 2. Dataset and data quality

The ZIP contained two separate CSV files, each with 250,000 rows. This report uses **ecommerce_customer_data_large.csv** as the primary dataset; the custom_ratios file was not merged because it contains different transaction records. The primary file has no exact duplicate rows and no missing values in the essential customer ID, purchase date, or purchase amount fields. The Returns field is missing in 47,382 rows (19.0%); return-rate calculations therefore use only records with a known 0/1 value. Customer Age and Age are identical. There is no separate transaction ID, so each row is treated as one transaction.

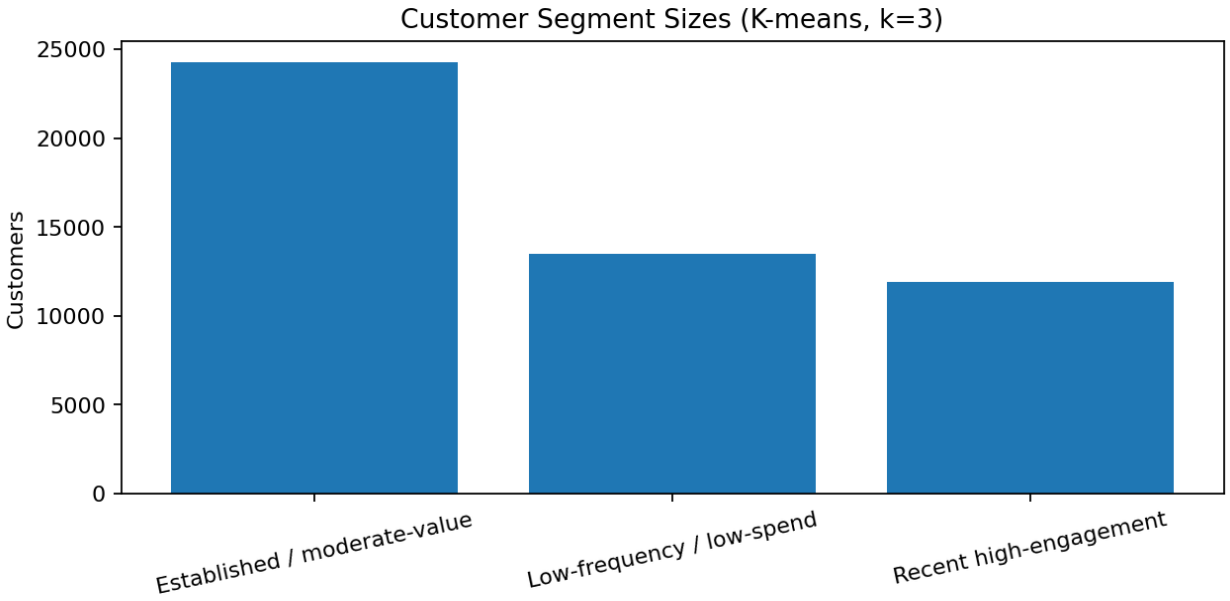
## 3. Customer behavior and RFM

RFM was calculated at customer level using snapshot date 14 Sep 2023. Average frequency is 5.03 transactions per customer and 96.7% of customers have more than one recorded transaction. The 90-day recently active share is 28.7%; 51.1% are over 180 days since last purchase and 25.4% are over 365 days.

## 4. K-means customer segments

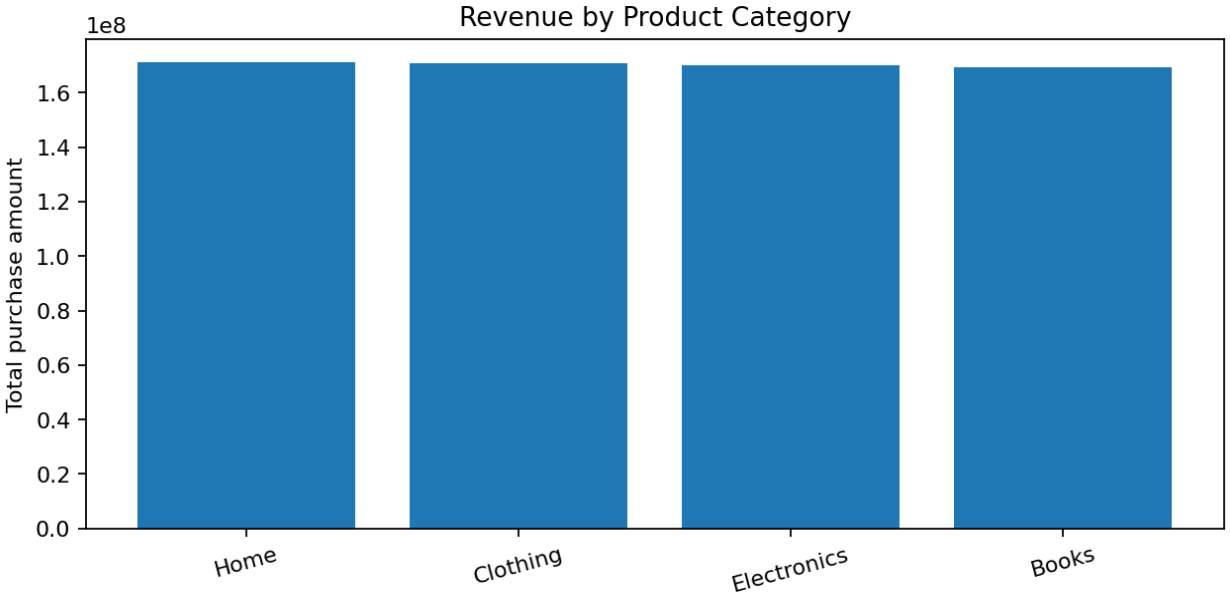
Three clusters were fitted to log-transformed and standardized Recency, Frequency, and Monetary values (random_state=42). Cluster IDs are arbitrary; the labels below are descriptive interpretations of the observed medians. The supplied churn rate is very similar across the clusters, so the segments should be used for engagement targeting rather than as a standalone churn classifier.

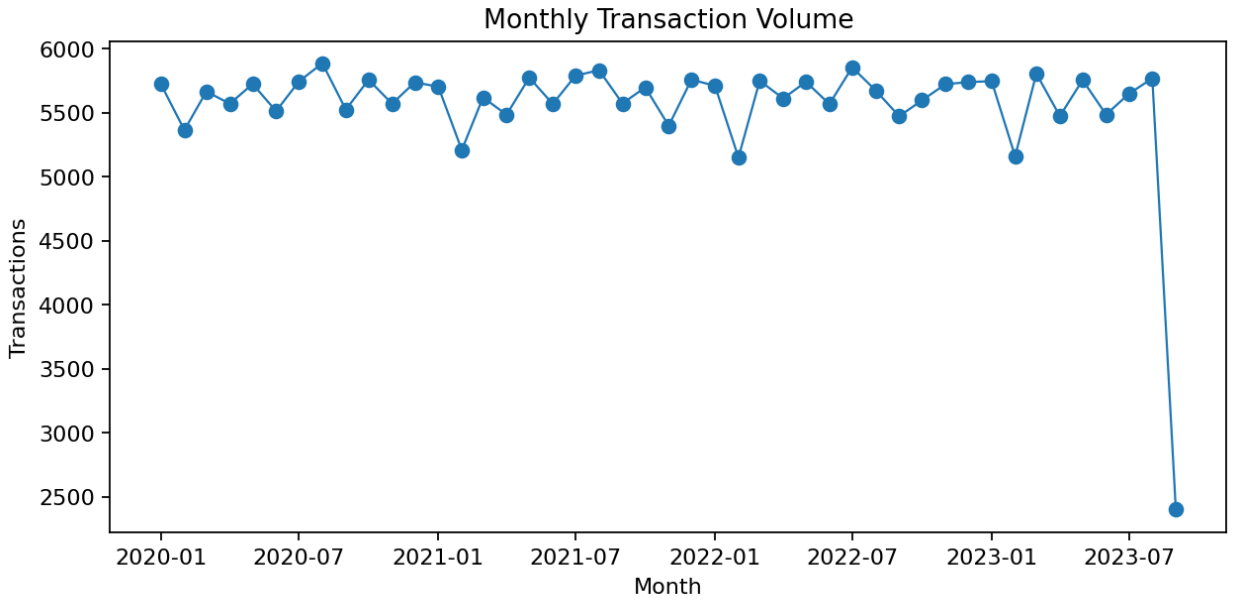
| Segment | Customers | Median recency (days) | Median frequency | Median monetary | Supplied churn |
|---|---|---|---|---|---|
| Established / moderate-value | 24,239 | 225 | 5 | 14,906 | 19.9% |
| Low-frequency / low-spend | 13,505 | 368 | 3 | 6,433 | 19.9% |
| Recent high-engagement | 11,917 | 36 | 6 | 16,975 | 20.3% |




## 5. Purchase patterns and retention

Revenue is broadly balanced across the four product categories. Home is the largest category by recorded purchase amount (171,138,916); the smallest is Books (169,345,236). Payment methods are also evenly distributed: Credit Card 33.4%, PayPal 33.4%, Cash 33.2%. Among rows where Returns is recorded, the return flag rate is 50.1%; this should be interpreted cautiously because almost one-fifth of rows have unknown return status.






Cohort analysis groups customers by first purchase month. Average observed month-2 retention among cohorts with sufficient follow-up is 10.8%; month-4 is 10.8%; month-7 is 10.2%. Later cohorts have less follow-up time, so cohort comparisons must account for right-censoring. Month 1 is 100% by definition.

## 6. Five actionable recommendations for Alfido Tech

**1. Launch inactivity-based win-back journeys.** Prioritize customers with 181–365 days since last purchase, then a separate dormant sequence for customers over 365 days. Use a relevant category reminder and a time-limited incentive; measure reactivation and incremental revenue against a holdout group.

**2. Improve the second-purchase conversion.** The dataset shows a large share of repeat customers, but one-time customers still form a meaningful group. Trigger a follow-up message after a first purchase, recommend complementary products, and test a small second-order benefit within 30 days.

**3. Create a loyalty path for recently active, high-value customers.** The recent high-engagement cluster has the shortest median recency and highest median spend. Offer early access, loyalty points, or personalized bundles to protect their repeat behavior without discounting every order.

**4. Use category-specific merchandising rather than broad discounts.** Category revenue is relatively balanced, so avoid assuming one category dominates customer demand. Test category-based bundles and cross-sell recommendations, then compare conversion and average order value by category.

**5. Fix return-data completeness and establish a retention dashboard.** Returns is unknown for 18.95% of transaction rows. Make return status mandatory at transaction close and track monthly cohort retention, recency bands, repeat purchase rate, average order value, and reactivation rate.

## 7. Limitations and interpretation

The dataset's currency is not identified, so monetary amounts are reported in raw dataset units. The supplied Churn label is constant within each customer in this file, but its business definition and labeling methodology are not documented. Inactivity thresholds are analyst-defined proxies. K-means is descriptive and does not establish causality. No promotional, channel, margin, or customer service data is available, limiting causal explanations and profitability analysis.

## Appendix: key metrics

| Metric | Result |
|---|---|

| | |
|---|---|
| Transactions | 250,000 |
| Unique customers | 49,661 |
| Recorded total purchase amount | 681,346,299 |
| Average transaction amount | 2,725.4 |
| Supplied churn label rate | 20.0% |
| Repeat-customer share | 96.7% |
| Returns missing | 19.0% |
| Known-return flag rate (known values only) | 50.1% |
| Customers inactive >180 days | 51.1% |